PROJECT BRIEF & TECHNICAL ARCHITECTURE

Workday Accounting Journal EIB Studio

Automated Desktop ETL Utility for Converting Raw ERP Ledger Exports into Workday-Ready EIB Workbooks (v45.1 – v48.0)

01 Why This Project Was Built & Problem Statement

Purpose: Organizations migrating financial data or posting periodic general ledger entries into **Workday Financials** must use Workday's *Enterprise Interface Builder (EIB)* Excel templates. This project was built to eliminate manual spreadsheet preparation by automatically transforming raw accounting exports (CSV/Excel) into validated, Workday-compliant EIB workbooks.

The Problem Statement (Manual Challenges)

- **Complex Multi-Column Schemas:** Workday EIB templates contain 30 to 96+ strict metadata columns where Rows 1–5 are locked system instructions and data begins strictly at Row 6.
- **Error-Prone Manual Mapping:** Manually copying debits, credits, header keys, and worktags across hundreds of rows frequently causes shifted columns or formula corruption.
- **Strict Worktag Formatting:** Workday rejects uploads if Cost Centers lose leading zeros (e.g., `90` instead of `090`), if Spend Categories are missing for specific Ledger Accounts, or if literal `"NULL"` text appears in memo cells.
- **Unbalanced Journals:** Even a $0.01 rounding discrepancy between total Debits and Credits causes the entire Workday EIB batch payload to fail.

Project Objectives

- **Zero-Touch ETL Automation:** Convert raw 2025 and 2026 accounting files into Workday EIB workbooks in under 2 seconds.
- **Multi-Version Template Support:** Support both legacy Two-Sheet (**v45.1 Import**) and modern Single-Sheet (**v46.0, v47.0, v48.0 Submit**) Workday EIB architectures.
- **Automated Reference Enrichment:** Automatically look up and populate *Spend Categories* from the master Ledger Account mapping table and zero-pad *Cost Centers* to 3 digits.
- **100% Standalone Desktop Execution:** Package everything—UI, Python ETL engine, and Excel templates—into a single offline `WorkdayEIBStudio.exe` file requiring zero installation.

02 Why We Found This App as the Optimal Solution

1. 100% Offline & Secure

Financial general ledger data is highly sensitive. Unlike cloud converters or web dashboards, this desktop app processes data strictly in local memory with zero external network calls or open ports.

2. Built-In Financial Guardrails

Before saving any workbook, the engine enforces blank integrity on `LineMemo` (stripping literal `"NULL"` strings), preserves template styles in Rows 1–5, and verifies $\sum ext{Debits} = \sum ext{Credits}$.

3. Zero Setup Portability

Finance team members do not need Python, Node.js, or template folders installed. Sharing `WorkdayEIBStudio.exe` gives any Windows user all 4 Workday EIB templates embedded inside the executable.

03 The Two Target EIB File Architectures: Import vs. Submit Accounting Journal

Workday provides two distinct EIB web service templates for loading accounting journals. Our application dynamically inspects the selected target template and switches its mapping engine automatically:

Feature / Aspect	1. Import Accounting Journal (v45.1)	2. Submit Accounting Journal (v46.0, v47.0, v48.0)
Template Structure	TWO-SHEET ARCHITECTURE Splits data across two relational tabs: • Sheet 1: `Import Accounting Journal` (Headers) • Sheet 2: `Journal Entry Line Replacement` (Lines)	SINGLE-SHEET ARCHITECTURE Consolidates header and line attributes into a single unified tab: • Sheet 1: `Submit Accounting Journal` (96 columns)
Relational Linking	Uses a parent-child `Header Key*` (Col B in Sheet 1) linked to `Header Key` (Col B in Sheet 2) and sequential `Line Key*` / `Line Order`.	Uses `Spreadsheet Key*` (Col B) for journal grouping, sets `Row ID* = 1` (Col AI), and sequences lines via `Line Order` (Col AJ).
Column Mapping Layout	**Sheet 1:** Company (Col L), Currency (Col M), Ledger Type (Col N), Date (Col P), Source (Col Q). **Sheet 2:** Account (Col F), Debit (Col J), Credit (Col K), Memo (Col T), Spend Cat (Col Y), Cost Center (Col AA).	**Unified Row (Row 6+):** Company (Col O), Currency (Col P), Ledger Type (Col Q), Date (Col S), Source (Col T), Account (Col AL), Debit (Col AP), Credit (Col AQ), Memo (Col AZ), Cost Center (Col BC), Spend Cat (Col BE).
Workday Workflow Purpose	Designed for high-volume batch imports where header metadata is normalized separately from hundreds of replacement journal lines.	Designed for direct submission into Workday business processes with auto-complete, submit flags (Col I), and dedicated worktag columns (v48.0).

04 Brief of Application Components

Component	Role / Layer	Technical Responsibilities
`WorkdayEIBStudio.exe`	Standalone Executable	Single-file Windows package compiled via PyInstaller. Bundles the Python interpreter, Pandas/OpenPyXL libraries, UI assets, lookup table, and all 4 EIB templates into `sys._MEIPASS`.
`desktop_app.py`	Native Desktop Bridge (PyWebView IPC)	Launches the native Windows WebView2 window and exposes the `DesktopApi` class directly to JavaScript (file/folder pickers, preset loaders, Excel launcher, and ETL triggers) without any HTTP server.
`frontend/index.html`	Desktop Workbench UI	Dark-themed desktop interface featuring one-click 2025/2026 presets, custom file picker, save-location selector, target template cards (v45.1–v48.0), KPI summary badges, and live terminal log output.
`run_agent.py`	Core ETL & Mapping Engine	Contains 21 specialized functions for source ingestion, header deduplication, Spend Category lookup, 3-digit Cost Center formatting, strict `LineMemo` sanitization, and Debit/Credit balancing.
`Spend Category Mapped Ledger Account.xlsx`	Reference Lookup Table	Master dictionary mapping `LedgerAccountReferenceID` codes to official Workday `Spend_Category_ID` worktags automatically during row processing.
`target/*.xlsx`	Embedded EIB Templates	Official Workday template workbooks (`v45.1` , `v46.0` , `v47.0` , `v48.0`) whose first 5 instructional header rows and formatting are preserved intact.

05 How the Project Works (End-to-End Internal Workflow)

1. Ingestion & Detection
Reads raw CSV or multi-sheet Excel (2025/2026). Normalizes headers and strips strings like `"NULL"` or `"N/A"` into clean Python `None` values.

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2. Template Inspection
Opens the chosen target EIB template (v45.1–v48.0), detects Two-Sheet vs. Single-Sheet mode, and clears old rows from Row 6 downward.

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3. Worktag Enrichment
Zero-pads Cost Centers to 3 digits (`000` , `090` , `200`), maps Spend Categories via Ledger Account lookup, and keeps empty `LineMemo` cells strictly blank.

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4. Balance & Export
Computes total Debits, Credits, and Net Variance ($\Delta = 0.00$). Saves with strict Workday naming to the chosen local folder.

Strict Standardized Output Naming Convention

Regardless of whether the user runs a single file or batch mode, the engine enforces standardized enterprise output filenames in the selected destination directory:

- **2025 Source File** (`Accounting_Journal_Raw_LE01_202501.xlsx`) → `Import_Accounting_Journal_v45.0_LE01_2025_P1.xlsx` (1 Header, 8 Replacement Lines, $1,510.00 Balanced)
- **2026 Source File** (`Accounting_Journal_Raw_LE01_202601.xlsx`) → `Import_Accounting_Journal_v45.0_LE01_2026_P2.xlsx` (15 Headers, 439 Replacement Lines, $415,577,315.48 Balanced)

06 How to Work with This App (Step-by-Step User Guide)

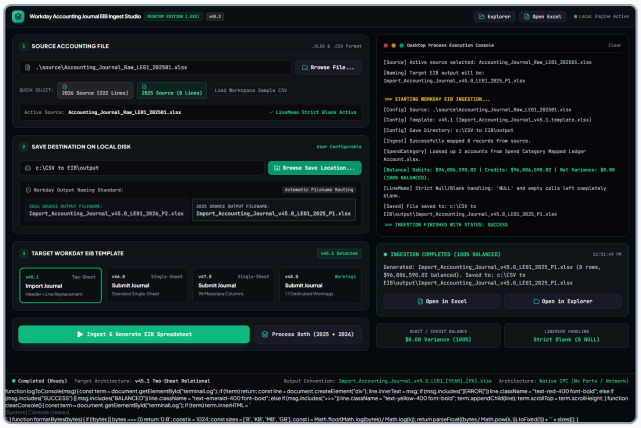


Figure 1: Workday EIB Studio Native Desktop Workbench showing Step-by-Step Controls & Live Verification Metrics

Step 1: Launch & Select Source File

Double-click `WorkdayEIBStudio.exe`. In **Step 1 (Source Accounting File)**, click one of the quick preset buttons (*2025 Source*, *2026 Source*, or *Batch Both*) or click **Browse Local File...** to pick any raw CSV/Excel export.

Step 2: Choose Target EIB Template (v45–v48)

In **Step 2 (Target EIB Template)**, select either **v45.1 Import Accounting Journal** (Two-Sheet) or **v46.0 / v47.0 / v48.0 Submit Accounting Journal** (Single-Sheet) based on your Workday integration type.

Step 3: Select Output Save Location

In **Step 3 (Output Destination)**, click **Browse Save Location...** to choose the local disk or network folder where the generated EIB Excel files should be written.

Step 4: Run Conversion & Open in Excel

Click **Run EIB Conversion**. Review the live Total Debits, Total Credits, and `BALANCED ($0.00)` badge on the right panel, then click **Open in Excel** or **Show in Folder** to upload directly to Workday.